Notes: Englander, Karp & Simon (RES, 2026)
The Value of Information in a Congested Fishery

Contents

1 Big picture

- **Question.** What is the value of public and private information when agents use information to choose locations in a congested common-resource setting?
- **Setting.** The empirical application is Peru's North-Central anchoveta fishery, the world's largest fishery. Firms choose where to fish using public signals from satellite ocean data and private signals from their own recent fishing experience.
- **Key tension.** More precise information helps firms locate closer to the densest fish stock, but it can also make firms crowd into the same location. The welfare value of information therefore depends on the strength of congestion.
- **Public versus private information.** Public information tends to coordinate everyone on the same location and can worsen congestion. Private information improves each firm's forecast while preserving more dispersion across firms.
- **Model.** Each firm receives a public signal y and a private signal x_i about the ideal fishing location θ. It chooses a location k_i trading off proximity to θ against congestion from being close to other firms.
- **Main structural estimates.** The paper estimates the ratio of public-to-private signal noise as $r = 0.923$, the BLUE weight on public information as $\delta = 0.540$, and the congestion parameter as $\tau = 0.448$.
- **Main welfare result.** At the estimated parameters, more precise private information substantially raises welfare, with elasticity about $\beta = 1.98$. More precise public information has a very small positive welfare effect, with elasticity about $\alpha = 0.02$.
- **Information clubs.** Exclusionary sharing of private information within clubs raises welfare in the estimated fishery, especially when the club contains a modest fraction of firms.
- **Global information sharing.** A regulator or association can collect a fraction of private information to improve the public signal. This can raise welfare when enough firms are active, but a small amount of sharing can reduce welfare when the market is thin.
- **Economic takeaway.** Information is not always more valuable when it is more public. In congested settings, the social value of information depends on whether it improves accuracy while preserving beneficial dispersion.

2 Institutional context and data measurement

2.1 Peruvian anchoveta fishery

The industry is a useful empirical setting because a large number of industrial vessels choose where to fish in a long, narrow coastal fishery where local congestion is plausible. The regulator sets a total allowable catch each season, so the paper focuses on within-season location choices and congestion rather than long-run stock depletion.

- The regulator, PRODUCE, sets an industry-wide total allowable catch after consulting IMARPE.
- There are two fishing seasons per year for Peru's North-Central anchoveta stock.
- Vessels are assigned individual quotas, but their location choices are decentralized.
- Local congestion can arise because nearby vessels deplete local anchoveta and physically interfere with one another.

Interpretation: The regulatory environment helps the model focus on location choice and within-season congestion. Because catch limits reduce the importance of inter-seasonal stock externalities, the main friction is where firms choose to fish given information.

2.2 Electronic logbook data

The data contain all industrial-vessel fishing sets in 2017–2019. A "set" is a vessel-level fishing operation observed at a precise time, longitude, and latitude.

- The data cover 806 industrial vessels.
- The sample contains 246,920 sets across six fishing seasons.
- For each set, the paper observes location, time, vessel identity, firm identity, and catch.
- The data allow the authors to construct fine spatial measures of catch per unit effort and congestion.

Interpretation: The unit of observation is very local: a vessel choosing a specific place to deploy its net. This makes the data unusually well suited to estimating a spatial congestion model.

2.3 Catch per unit effort (CPUE)

The paper constructs a measure of CPUE by residualizing tons caught per set on vessel and firm characteristics:

$$CPUE_{ift} = \text{tons per set}_{ift} - \widehat{E}[\text{tons per set}_{ift} \mid \text{vessel and firm traits}].$$

Controls include vessel length, engine horsepower, gross tonnage, number of vessels owned by the firm, and firm-type indicators.

Interpretation: CPUE is the empirical proxy for local anchoveta density net of fishing cost differences. A high residual means a location generated unusually high catch after adjusting for vessel and firm capacity.

2.4 Fishing zones and ideal location

Each set occurs in one of 90 regulatory fishing zones. The paper uses zone-level CPUE to proxy for the daily ideal fishing ground. Because daily true biomass is unobserved, the daily best zone is inferred from catch outcomes.

- The regulator defines 90 zones.
- The paper constructs polygon boundaries from observed sets in each zone.
- Zone-level average CPUE is used as a less noisy proxy for biomass than the single best set.
- Distance to the ideal zone is used to estimate the cost of missing the best fishing ground.

Interpretation: The empirical model converts a two-dimensional coastline into a location-choice problem. The zone construction is a practical bridge between the one-dimensional theoretical model and the two-dimensional fishery.

2.5 Public information

Public information consists of daily satellite-based oceanic variables from NASA and NOAA:

- chlorophyll;
- sea surface temperature;
- sea surface temperature anomaly;
- sea surface salinity;
- sea level anomaly.

Multiple satellites yield nine public signal variables before transformations and interactions.

Interpretation: These data are public because all firms can access the same satellite information. Public signals can help locate fish but may also coordinate firms into the same places.

2.6 Private information

Private information is constructed from the previous day's CPUE and locations of the same firm's vessels. For each set, the paper uses functions of prior CPUE and prior locations within the firm, including nearest previous-day sets and pairwise interactions.

Interpretation: Private information is firm-specific because it comes from the firm's own recent fishing experience. It can improve a firm's location choice without making all firms move in the same direction.

3 Model and theoretical specifications

3.1 Signals

There are $n < \infty$ firms. Nature chooses the ideal location θ. Each firm receives a public signal and a private signal:

$$y = \theta + \varepsilon_y, \qquad \varepsilon_y \sim N(0, \sigma_y^2),$$
$$x_i = \theta + \varepsilon_{x_i}, \qquad \varepsilon_{x_i} \sim N(0, \sigma_x^2).$$

Private-signal noise is uncorrelated across firms, while private and public signal noise may be correlated:

$$E[\varepsilon_{x_i}\varepsilon_{x_j}] = 0 \quad (i \neq j), \qquad E[\varepsilon_y \varepsilon_{x_i}] = \rho\sigma_y\sigma_x.$$

Interpretation: The model separates two information channels. Public information is common and therefore affects coordination. Private information is dispersed and therefore preserves heterogeneity in actions.

3.2 BLUE weight and relative precision

Define the ratio of standard deviations:

$$r \equiv \frac{\sigma_y}{\sigma_x}.$$

A larger r means the public signal is relatively less precise. The BLUE weight on the public signal is

$$\delta = \frac{1 - \rho r}{1 + r^2 - 2\rho r}.$$

When $\rho = 0$, this simplifies to

$$\delta = \frac{1}{1 + r^2}.$$

Interpretation: δ is a statistical benchmark for how much weight a best linear estimator of the ideal location places on the public signal. It is not necessarily the equilibrium weight because strategic congestion changes location incentives.

3.3 Payoff and congestion

If firm i chooses location k_i and the other firms choose locations k_j, firm i's payoff is

$$J(k_i \mid \{k_j\}, \theta) = \frac{1}{n-1}\frac{B}{2}\sum_{j\neq i}(k_i - k_j)^2 - \frac{A}{2}(k_i - \theta)^2.$$

The parameter $A > 0$ is the cost of being far from the ideal fishing location. The parameter B captures the payoff from dispersion. Define the normalized congestion parameter:

$$\tau \equiv \frac{B}{A}.$$

Interpretation: When $\tau > 0$, firms benefit from being far from other firms because local congestion is costly. When $\tau < 0$, proximity has positive spillovers. The empirical fishery is estimated to have $\tau > 0$.

3.4 Linear decision rules and information clubs

With linear strategies, a firm chooses a location using a weight on the public signal:

$$k_i = \gamma y + (1-\gamma)x_i.$$

If firms form information clubs of size c, each club member observes the average private signal of its club. The Bayesian Nash equilibrium weight on the public signal is

$$\gamma^{NE,c} = \frac{1 - \frac{n-c}{n-1}\tau - cr\rho}{1 - \frac{n-c}{n-1}\tau + cr^2 - 2cr\rho}.$$

Interpretation: The equilibrium weight differs from the statistical BLUE weight because firms care not only about estimating θ, but also about how their locations co-move with other firms.

3.5 Equilibrium payoff

The per-firm equilibrium payoff can be written as

$$P = \frac{A}{2}\sigma_x^2 S(\tau, \gamma(r, \tau, \rho, c), r, \rho, c),$$

where $S(\cdot)$ is a normalized payoff function. The paper uses this expression to derive welfare effects of changing information precision or reallocating information.

Interpretation: The payoff expression turns the qualitative tradeoff into a parameterized welfare calculation. Once r and τ are estimated, the theory can be taken to data.

3.6 Exogenous changes in precision

With no clubs ($c = 1$) and $\rho = 0$, the welfare effects of more precise public and private information depend on the point (τ, δ).

For public information, the key boundary is

$$\delta_a(\tau) = \frac{3\tau - 1}{\tau^2 + \tau}.$$

For private information, the key boundary is

$$\delta_b(\tau) = \frac{2\tau - 1}{\tau}.$$

Interpretation: Figure 1 maps the regions where more precise public or private information raises welfare. Public precision is more likely to be harmful because it coordinates firms more tightly, increasing congestion.

3.7 Global information sharing

A regulator may collect a fraction f of firms' private information and turn it into a more precise public signal. Let ϖ be the fractional reduction in the variance of the public signal. The new moments are

$$\sigma_y^{2\prime} = (1-\varpi)\sigma_y^2,$$

$$\sigma_x^{2\prime} = \frac{1}{1-f}\sigma_x^2,$$

$$r' = r\sqrt{(1-\varpi)(1-f)} \leq r.$$

Interpretation: Global information sharing improves the public signal but degrades private signals. The welfare effect depends on whether the gain from a better public signal exceeds the loss of private precision and the extra congestion from common information.

4 Identification and empirical design

4.1 What is identified

The paper estimates two primitives that determine the value of information:

$$r = \frac{\sigma_y}{\sigma_x}, \qquad \tau = \frac{B}{A}.$$

The first measures the relative noise in public and private signals. The second measures how valuable dispersion is relative to being near the best fishing ground.

Interpretation: The empirical design is structural: the authors do not estimate a treatment effect of an information policy directly. Instead, they estimate model parameters and then use the model to compute counterfactual welfare effects.

4.2 Identification of public and private signal precision

The public signal variance σ_y^2 is estimated by predicting the squared distance from a set's location to the best local zone using public satellite variables. The private signal variance σ_x^2 is estimated with the same procedure, replacing public predictors with firm-specific previous-day information.

The core prediction equation is

$$\left\|(k_{ifzt} - \widehat{\theta}_{izt})^b\right\|^2 = \psi X_{ifzt} + e_{ifzt},$$

where X_{ifzt} is either a matrix of public signal variables or private signal variables.

Interpretation: The residual sum of squares is interpreted as signal noise. If public predictors explain the ideal location better, the public signal is more precise; if private predictors explain it better, private information is more precise.

4.3 Machine-learning implementation

The paper uses lasso with a training-test split:

- 75% of days are randomly assigned to the training set;
- the remaining days form the test set;
- 10-fold cross-validation chooses the lasso penalty;
- prediction error in the test set is used to estimate signal variance.

Interpretation: The out-of-sample test set matters because the public and private predictor matrices are large. Without out-of-sample validation, overfitting could falsely make a signal appear precise.

4.4 Identification of congestion parameter τ

The paper estimates τ from how CPUE varies with dispersion and distance to the best local zone:

$$CPUE_{izt} = \kappa + BD_{it} + A \left\| (k_{izt} - \widehat{\theta}_{izt})^b \right\|^2 + e_{izt}.$$

The paper estimates

$$\tau = \frac{B}{-A}.$$

Interpretation: Dispersion raises CPUE if local congestion is costly. Distance to the best zone lowers CPUE because firms are farther from high biomass. The ratio translates these two effects into the normalized congestion parameter.

4.5 Threats to identification

- **Unobserved biomass.** High biomass may raise CPUE and affect vessel clustering. The paper addresses this with post-double-selection controls.
- **Mechanical distance relationship.** Distance to the best local zone is partly mechanical because the best zone is inferred from CPUE. The dispersion coefficient is the more important congestion component.
- **Signal correlation.** The estimated correlation between public and private errors is large relative to the positive-definiteness bound for the median number of firms. The main analysis sets $\rho = 0$ and checks robustness.
- **Two-dimensional fishing choice.** The model is one-dimensional, while the fishery is spatially two-dimensional. The long coastal shape of the fishery makes the approximation plausible, and extensions discuss two-dimensional choices.
- **CPUE as biomass proxy.** CPUE is a standard proxy for local biomass, but it also depends on effort and vessel characteristics. The paper residualizes CPUE to reduce this concern.

Interpretation: The main identification strength is that the theory maps observable spatial behavior into a small number of parameters. The main limitation is that true biomass and true information sets are not directly observed.

4.6 Bottom line

The empirical design links data to theory in three steps:

1. Estimate public and private signal precision from out-of-sample predictive performance.

2. Estimate congestion from the effect of spatial dispersion on CPUE relative to the cost of missing the best zone.
3. Use the structural model to compute welfare effects of better public information, better private information, information clubs, and global information sharing.

Interpretation: The core design is not a reduced-form policy experiment. It is a structural measurement exercise that turns rich fishing-location data into welfare-relevant parameters.

5 Tables and figures

5.1 Figure 1: Boundaries that determine the welfare effect of more precise information

Read:

- The horizontal axis is congestion, τ; the vertical axis is the BLUE weight on public information, δ.
- Curve a is the public-information boundary $\delta_a(\tau)$.
- Curve b is the private-information boundary $\delta_b(\tau)$.
- More precise public information raises welfare only above curve a.
- More precise private information raises welfare only above curve b.
- The point X marks the preferred estimate $(\tau, \delta) = (0.448, 0.540)$.

Detailed interpretation: The key visual message is that public information and private information have different welfare thresholds. The region between curves a and b is especially important: in that region, private precision is valuable but public precision is harmful. This happens because public information coordinates actions more strongly and can intensify congestion.

How it fits the paper: The estimated point lies in the region where both types of precision improve welfare, but the point is near the public-information boundary. This explains why the estimated public-information elasticity is tiny, while the private-information elasticity is large.

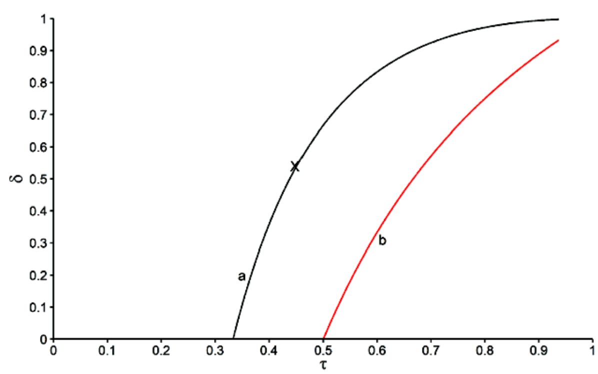

Figure 1: Boundaries that determine the welfare effect of more precise information. The figure is the paper's theoretical map from primitives to welfare signs. It shows why "more information" is not always better in a congested environment and why private information can be more socially valuable than public information.

5.2 Figure 2: Peruvian electronic logbook data, 2017–9; Figure 3: Fishing zone polygons. (a) 2017–9. (b) 24 May 2018

Read:

- Figure 2 plots each vessel-level fishing set and colors it by CPUE.
- The logbook data cover 246,920 sets by 806 unique vessels.
- Figure 3(a) converts observed sets into convex-hull polygons for the 90 regulatory fishing zones.
- Figure 3(b) shows daily zone boundaries for one example day, illustrating how daily polygons are constructed from active sets.

Detailed interpretation: These figures translate the economic model into the data. The model assumes location choice; the figures show that the paper observes location choices at high spatial resolution. The figures also show that fishing is concentrated close to the coast, making the one-dimensional approximation along the coastline more plausible.

How it fits the paper: Figure 2 motivates CPUE as the empirical outcome and Figure 3 motivates the best-zone and distance measures. Without these spatial data, the paper could not estimate either signal precision or congestion.

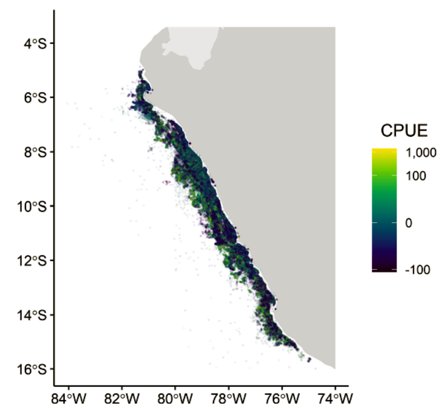

Figure 2: Peruvian electronic logbook data, 2017–9. Each point is a set and color represents residualized CPUE.

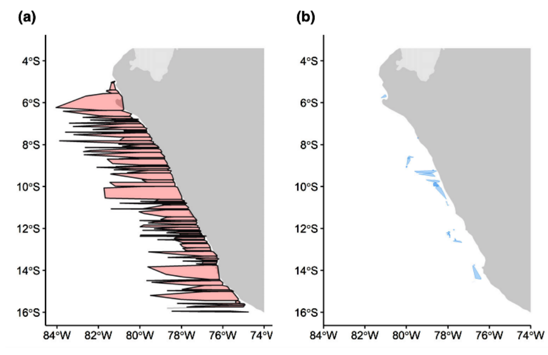

Figure 3: Fishing zone polygons. Panel (a) uses all 2017–9 data; panel (b) shows one daily example.

Figures 2 and 3: Data construction and spatial measurement. These figures show how the authors observe set-level location and convert the observed fishing geography into zone-level measures of CPUE, ideal location, and distance.

5.3 Table 1: Public and private signal parameter estimates; Table 2: Estimates of τ

Read:

- Table 1 reports $\sigma_y^2 = 9.19956 \times 10^{11}$ and $\sigma_x^2 = 1.07994 \times 10^{12}$.
- The estimated ratio is $r = 0.923$, meaning the public signal is slightly more precise than the private signal.
- Setting $\rho = 0$ gives $\delta = 0.540$, so the BLUE places about 54% weight on the public signal.
- Table 2 estimates dispersion and distance effects on CPUE.
- Column 1 gives $B = 1.596$, $A = -3.558$, and $\tau = 0.448$.
- Column 2 adds post-double-selection controls; τ becomes smaller but remains qualitatively consistent with positive congestion.

Detailed interpretation: Table 1 pins down the relative informativeness of public and private signals. Table 2 pins down the severity of congestion. Together, the two tables locate the fishery on the theoretical map in Figure 1. The model's welfare conclusions depend on both: if private information were much less precise or congestion much larger, the welfare signs could change.

How it fits the paper: These are the core calibration inputs. The later policy counterfactuals–private precision, public precision, clubs, and global sharing–all use these estimates.

TABLE 1
Public and private signal parameter estimates

σ_y^2	σ_x^2	r	δ
9.19956e+11	1.07994e+12	0.923	0.540

(a) Table 1: Public and private signal parameter estimates.

TABLE 2
Estimates of τ

	CPUE	
	(1)	(2)
Dispersion	1.596	0.496
	(0.954)	(1.013)
DistToBest^2	−3.558	−3.894
	(0.892)	(0.734)
τ	0.448	0.127
	(0.265)	(0.259)
Controls	No	Yes
Adj. R^2	0.005	0.066
N	246,920	246,920

Notes: The dependent variable is catch per unit effort (CPUE). Mean CPUE is 0 by construction; the standard deviation is 51.8. We estimate τ as the Dispersion coefficient divided by the negative of the DistToBest^2 coefficient. We estimate the standard error of τ with the delta method. Our post-double-selection procedure retains 265 control variables that predict CPUE or Dispersion (Column 2). We two-way cluster standard errors by date and zone.

(b) Table 2: Estimates of τ.

Tables 1 and 2: Structural parameter estimation. The first table estimates information precision; the second estimates the congestion parameter from CPUE, dispersion, and distance to the best zone.

5.4 Figure 4: Normalized payoff as a function of club size

Read:

- The horizontal axis is ϕ, the fraction of other firms that belong to firm i's information club.
- The figure shows normalized payoff for $n = 38$, $n = 135$, and $n = 234$, the 25th, 50th, and 75th percentiles of daily active firms.
- Payoff rises sharply as ϕ increases from 0 to roughly 0.15.
- Further increases in club size generate much smaller gains.

Detailed interpretation: Information clubs make private information more precise within the club while preserving some separation from nonmembers. This helps firms forecast the ideal location without making the whole industry coordinate on one signal. The diminishing gains after modest club sizes reflect the fact that averaging a small set of private signals already produces a large precision improvement.

How it fits the paper: The figure supports the policy idea that exclusionary information sharing may raise welfare. It also highlights a tradeoff: clubs can improve information, but the cost and feasibility of forming them matter.

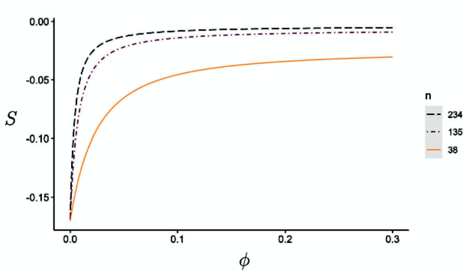

Figure 4: Normalized payoff as a function of club size. The graph shows that moderate information clubs can substantially raise welfare, while very large clubs add little incremental benefit.

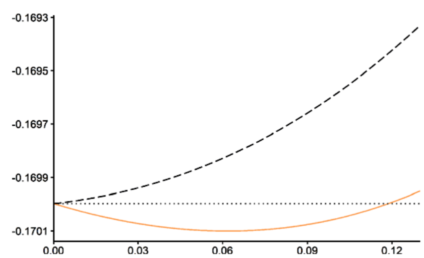

FIGURE 5

Figure 5: Normalized payoff as a function of global information sharing. The figure shows a threshold effect: extensive global sharing can be valuable, but small-scale sharing can reduce welfare when the number of active firms is small.

5.5 Figure 5: Normalized payoff as a function of global information sharing

Read:

- The horizontal axis is ϖ, the reduction in the variance of the public signal due to global information sharing.
- The vertical axis is normalized payoff $S/(1-f)$.
- For $n = 234$, welfare increases with global information sharing.
- For $n = 38$, a small amount of global sharing lowers welfare; welfare rises only once $\varpi > 0.119$.

Detailed interpretation: Global information sharing simultaneously improves the public signal and degrades private information. With many firms, the regulator can improve the public signal by taking only a small amount of private information from each firm, so sharing is beneficial. With few firms, the loss of private information can dominate unless sharing is large enough.

How it fits the paper: Figure 5 shows why policy advice depends on market thickness. The same information-sharing rule can be harmful in thin markets and beneficial in thick markets.

6 Short summary

- The paper studies the value of information in a congested fishery.
- Public information and private information both help firms locate dense fish stocks, but public information coordinates firms more strongly and can increase congestion.
- The model has firms choose locations based on public and private signals about the ideal fishing ground.
- Welfare depends on relative signal precision r, the public-signal BLUE weight δ, and congestion severity τ.
- The data consist of 246,920 fishing sets by 806 industrial vessels in Peru's anchoveta fishery from 2017 to 2019.
- Public signals are satellite ocean variables; private signals are firm-specific previous-day CPUE and locations.
- The estimated public signal is slightly more precise than the private signal, with $r = 0.923$ and $\delta = 0.540$.
- The estimated congestion parameter is $\tau = 0.448$.
- More precise private information has a large positive welfare effect; more precise public information has a negligible welfare effect.
- Information clubs raise welfare, especially when clubs are modest in size.
- Global information sharing can raise welfare when many firms are active, but small sharing can reduce welfare in thin markets.
- The main policy message is that information policies should account for congestion and for whether information is public or private.

7 Conclusion

7.1 Core conclusion

The paper's core conclusion is that information has a congestion-dependent social value. More precise private information is strongly beneficial in the estimated fishery, whereas more precise public information has a negligible welfare effect. Public information coordinates firms, and coordination can increase crowding. Private information improves individual targeting while leaving firms more dispersed.

7.2 Economic intuition

The intuition is that there are two margins. First, better information reduces the distance between a firm and the densest fish stock. Second, better information can reduce dispersion across firms. Public information affects both firms and their rivals in the same way, so it tends to make vessels move together. Private information is partly idiosyncratic, so it improves targeting without causing as much bunching. Congestion turns this distinction into a welfare result.

7.3 Policy implications

The results imply that regulators and industry associations should be cautious about making all information public. Better public measurement may not raise welfare much if it increases crowding. By contrast, tools that improve firms' private forecasting ability may be more valuable. The information-club results suggest that limited sharing among subsets of firms can increase welfare, while the global-sharing results show that broad information pooling is more attractive when many firms are active.

7.4 Interpretation of the empirical estimates

The estimates place the anchoveta fishery in a region where congestion is meaningful but not so severe that better information is generally harmful. The private-information elasticity of welfare is almost 2, while the public-information elasticity is close to zero. Therefore, the paper's quantitative message is not merely that public and private information differ in sign; it is that their magnitudes differ sharply.

7.5 Limitations

The model abstracts from some features of the fishery, including two-dimensional strategic choice, dynamic stock externalities, and heterogeneous firms. The data do not observe true biomass or true firm information sets, so the paper uses CPUE and predictive models as proxies. These limitations matter, but the paper addresses them through extensions, robustness analysis, and the use of high-resolution spatial data.

7.6 Takeaway

The paper changes the usual intuition that better information is always good. In congested environments, the type and distribution of information matter. Information that improves accuracy while preserving dispersion can be highly valuable; information that coordinates everyone into the same action may have much lower value or can even reduce welfare.